

Zebra TC57 Hardware Repair 001

Documentation

Device Information

Device Type: Zebra TC57 Mobile Computer

Department/Location: Enterprise Operations

Repair Type: Hardware Repair – LCD Display Replacement

Technician: Ridley Lherisse

Date Received: 09/26/2026

Date Completed: 09/29/2026

Issue Reported

Problem Description:

The Zebra TC57 device was received with a damaged display assembly. The device powered on and scanner functionality remained operational; however, the display showed a grey screen with no visible graphics output.

Initial Symptoms:

- Display was damaged/unusable
 - Device functionality needed to be evaluated
 - Hardware inspection required to determine whether the issue was limited to the display assembly or involved internal components
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Troubleshooting Process

Initial Assessment

1. Confirmed the device condition before beginning repair.
2. Powered down the device and disconnected power source.

3. Inspected the exterior housing for additional damage.
 4. Evaluated whether the issue appeared related to:
 - LCD display assembly
 - Display connector/flex cable
 - Internal motherboard components
 - Other hardware failures
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Repair Procedure

Device Disassembly

1. Powered down & removed battery of the Zebra TC57 and prepared the device for hardware service.
 2. Applied 91% isopropyl alcohol in controlled amount around the edges of the display screen to help loosen the adhesive bond and safely separate the screen from the device housing
 3. Carefully separate the device display screen from the housing using appropriate electronics repair tools while avoiding unnecessary pressure on internal components, connectors and cables.
 4. Removed the damaged display of assembly and inspected the internal housing area of any signs of
 - Physical Damage
 - Loose connections
 - Damaged flex cables
 - Debris or contamination
 - Potential damage to motherboard components or chips
 5. Before installing the replacement display, I inspected the housing, motherboard area, connectors, and surrounding components to confirm the issue was isolated to the display assembly and that no additional damage was present.
 6. Cleaned the repair area by removing broken glass fragments and leftover adhesive residue. Used controlled cleaning methods such as a cotton swab with the 91% iso alcohol, helpful with removing dust, fingerprints, oils, glass fragments & residue.
 7. Installed the replacement LCD/display assembly, ensuring proper alignment and secure connection of all components.
 8. Reconnected the battery, wait 30 seconds to a minute. Powered on the zebra device. Tested the touchscreen, brightness and display connections.
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Internal Inspection

Performed inspection of internal components before installing replacement parts.

Inspected:

- Motherboard condition
- Display connector
- Flex cable connection
- Internal chips/components
- Signs of physical damage or debris

Inspection Result:

No additional visible damage was identified. Internal components appeared suitable for replacement display installation.

Replacement Installation

1. Installed replacement LCD/display assembly.
 2. Ensured the display assembly was properly aligned and seated.
 3. Verified internal connections were secure.
 4. Reconnected battery after completing installation.
 5. Allowed device components to stabilize before powering on.
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Testing and Verification

After repair completion, the following tests were performed:

- ✓ Device powered on successfully
 - ✓ Display initialized correctly
 - ✓ Touchscreen responsiveness verified
 - ✓ Internal connections checked
 - ✓ Device functionality returned to normal operation
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Final Result

Repair Status: Completed Successfully

The Zebra TC57 was restored to normal working condition after replacing the damaged display assembly.

The repair process was documented to provide a clear troubleshooting history and support future maintenance procedures.

Lessons Learned / Notes

- Hardware issues should be diagnosed before replacing components.
- Proper inspection of connectors and internal components helps prevent repeat failures.
- Documentation improves troubleshooting efficiency and provides a repair history for future technicians.
- Verification testing after repair is essential before returning equipment to operation.